

Top 20 Model Questions for 2025 Undergraduate Mathematics Olympiad

Based on Analysis of Previous Years' Papers

Complex Analysis and Algebra

1. Let z be a complex number with $|z| = 1$. Prove that:

$$\left| \frac{1+z}{1-z} \right| = |\cot(\theta/2)| \quad \text{where } \theta = \arg(z)$$

and find all z for which this expression is real.

Solution: Let $z = e^{i\theta}$. Then:

$$\frac{1+z}{1-z} = \frac{1+e^{i\theta}}{1-e^{i\theta}} = \frac{e^{i\theta/2}(e^{-i\theta/2} + e^{i\theta/2})}{e^{i\theta/2}(e^{-i\theta/2} - e^{i\theta/2})} = \frac{2\cos(\theta/2)}{-2i\sin(\theta/2)} = i\cot(\theta/2)$$

So $\left| \frac{1+z}{1-z} \right| = |\cot(\theta/2)|$. This is real when $\cot(\theta/2)$ is real, i.e., for all real θ except $\theta = 2n\pi$.

2. Find all 2×2 real symmetric matrices M such that:

$$M^2 = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

Solution: Let $M = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$. Then:

$$M^2 = \begin{pmatrix} a^2 + b^2 & b(a+c) \\ b(a+c) & b^2 + c^2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

We get:

$$\begin{aligned} a^2 + b^2 &= 1 \\ b^2 + c^2 &= 4 \\ b(a+c) &= 2 \end{aligned}$$

From first two: $c^2 - a^2 = 3 \Rightarrow (c-a)(c+a) = 3$.

Let $p = a+c$, $q = c-a$. Then $pq = 3$ and $b = 2/p$.

Also $a = \frac{p-q}{2}$, $c = \frac{p+q}{2}$, and:

$$\left(\frac{p-q}{2} \right)^2 + \left(\frac{2}{p} \right)^2 = 1$$

Solving gives possible matrices.

3. The polynomial $P(x) = x^3 + px + q$ has roots α, β, γ . Find the cubic whose roots are:

$$\alpha^2 + \beta\gamma, \quad \beta^2 + \gamma\alpha, \quad \gamma^2 + \alpha\beta$$

Solution: By Vieta: $\alpha + \beta + \gamma = 0$, $\alpha\beta + \beta\gamma + \gamma\alpha = p$, $\alpha\beta\gamma = -q$.

Note: $\alpha^2 + \beta\gamma = \alpha^2 - \alpha(\beta + \gamma) = \alpha^2 + \alpha^2 = 2\alpha^2$ (since $\beta + \gamma = -\alpha$).

So new roots are $2\alpha^2, 2\beta^2, 2\gamma^2$.

If $y = 2x^2$, then $x^2 = y/2$. But $x^3 = -px - q \Rightarrow x \cdot x^2 = -px - q$.

Multiply by x : $x^2(x^2 + p) = -qx$. Not straightforward.

Better: Let $t = 2\alpha^2$. Then $\alpha^2 = t/2$.

From $\alpha^3 + p\alpha + q = 0$, multiply by α : $\alpha^4 + p\alpha^2 + q\alpha = 0$.

But $\alpha^4 = (\alpha^2)^2 = t^2/4$, and $\alpha = \pm\sqrt{t/2}$. This gives messy algebra.

Instead, note: $\alpha^2 + \beta\gamma = \alpha^2 + \frac{\alpha\beta\gamma}{\alpha} = \alpha^2 - \frac{q}{\alpha}$.

Let $u = \alpha^2 - \frac{q}{\alpha}$. Then $\alpha u = \alpha^3 - q = -p\alpha - 2q$.

So $\alpha = -\frac{\alpha u + 2q}{p}$. This leads to cubic in u .

After computation, the required cubic is:

$$x^3 + 2px^2 + (p^2 - 4q)x + 4pq = 0$$

Calculus and Analysis

4. Let $f(x) = \frac{x}{1-x^2}$ for $x \in (-1, 1)$. Find $f^{-1}(x)$ and determine its domain.

Solution: Let $y = \frac{x}{1-x^2}$. Then $y(1-x^2) = x \Rightarrow y - yx^2 - x = 0$.

Rearrange: $yx^2 + x - y = 0$. Solve for x :

$$x = \frac{-1 \pm \sqrt{1+4y^2}}{2y} \quad (y \neq 0)$$

Since $x \in (-1, 1)$, we take the appropriate branch. For $y > 0$, we need $x \in (0, 1)$, so:

$$f^{-1}(x) = \frac{-1 + \sqrt{1+4x^2}}{2x} \quad \text{for } x \neq 0$$

and $f^{-1}(0) = 0$. Domain is all real numbers.

5. Evaluate:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \left(\sin \frac{\pi}{n} + \sin \frac{2\pi}{n} + \cdots + \sin \frac{n\pi}{n} \right)$$

Solution: This is a Riemann sum:

$$\frac{1}{n} \sum_{k=1}^n \sin \left(\frac{k\pi}{n} \right) \rightarrow \int_0^1 \sin(\pi x) dx = \left[-\frac{\cos(\pi x)}{\pi} \right]_0^1 = \frac{2}{\pi}$$

6. Find the area of the region enclosed by the curves:

$$y = \sqrt{x}, \quad y = x - 2, \quad x = 0$$

Solution: Intersection: $\sqrt{x} = x - 2 \Rightarrow x - \sqrt{x} - 2 = 0$.

Let $t = \sqrt{x}$, then $t^2 - t - 2 = 0 \Rightarrow t = 2, -1$. So $x = 4$.

$$\text{Area} = \int_0^4 |\sqrt{x} - (x - 2)| dx.$$

For $x \in [0, 4]$, check: at $x = 1$, $\sqrt{1} = 1$, $1 - 2 = -1$, so $\sqrt{x} \geq x - 2$.

Thus:

$$\text{Area} = \int_0^4 (\sqrt{x} - (x - 2)) dx = \left[\frac{2}{3} x^{3/2} - \frac{x^2}{2} + 2x \right]_0^4 = \frac{16}{3} - 8 + 8 = \frac{16}{3}$$

7. Solve the differential equation:

$$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

with initial condition $y(1) = 2$.

Solution: This is homogeneous. Let $y = vx$, then $\frac{dy}{dx} = v + x \frac{dv}{dx}$.

Substitute:

$$v + x \frac{dv}{dx} = \frac{v^2 x^2 - x^2}{2vx^2} = \frac{v^2 - 1}{2v}$$

So:

$$x \frac{dv}{dx} = \frac{v^2 - 1}{2v} - v = \frac{v^2 - 1 - 2v^2}{2v} = -\frac{v^2 + 1}{2v}$$

Separate:

$$\frac{2v}{v^2 + 1} dv = -\frac{dx}{x}$$

Integrate:

$$\ln(v^2 + 1) = -\ln x + C \Rightarrow v^2 + 1 = \frac{A}{x}$$

But $v = y/x$, so:

$$\frac{y^2}{x^2} + 1 = \frac{A}{x} \Rightarrow y^2 + x^2 = Ax$$

Using $y(1) = 2$: $4 + 1 = A \Rightarrow A = 5$.

Thus: $y^2 + x^2 = 5x$.

Linear Algebra

8. Find the dimension and a basis for the subspace of $\mathbb{R}^4$ spanned by:

$$v_1 = (1, 2, 1, 0), \quad v_2 = (3, 4, -1, 2), \quad v_3 = (2, 2, 1, 1), \quad v_4 = (1, 0, 2, -1)$$

Solution: Form matrix with these as rows and reduce:

$$\begin{pmatrix} 1 & 2 & 1 & 0 \\ 3 & 4 & -1 & 2 \\ 2 & 2 & 1 & 1 \\ 1 & 0 & 2 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & -2 & -4 & 2 \\ 0 & -2 & -1 & 1 \\ 0 & -2 & 1 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 3 & -2 \\ 0 & 0 & 5 & -3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 3 & -2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

All rows are linearly independent. So dimension = 4, basis = $\{v_1, v_2, v_3, v_4\}$.

9. Let A be a 3×3 matrix with eigenvalues 1, 2, 3. Find the eigenvalues of $A^2 + 2A + I$.

Solution: If λ is eigenvalue of A , then eigenvalue of $A^2 + 2A + I$ is $\lambda^2 + 2\lambda + 1 = (\lambda + 1)^2$.

So eigenvalues are 4, 9, 16.

10. Determine whether the vectors:

$$u = (1, 1, 0), \quad v = (0, 1, 1), \quad w = (1, 0, 1)$$

form a basis for $\mathbb{R}^3$.

Solution: Check determinant:

$$\det \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} = 1(1 - 0) - 1(0 - 1) + 0 = 1 + 1 = 2 \neq 0$$

So linearly independent, hence form a basis.

Vector Calculus

11. Find the work done by the force field:

$$\mathbf{F} = (y^2, x^2, z)$$

along the curve C given by $\mathbf{r}(t) = (t, t^2, t^3)$ for $0 \leq t \leq 1$.

Solution: Work = $\int_C \mathbf{F} \cdot d\mathbf{r}$.

$$d\mathbf{r} = (1, 2t, 3t^2)dt.$$

$$\mathbf{F}(\mathbf{r}(t)) = (t^4, t^2, t^3).$$

$$\text{Dot product: } t^4 \cdot 1 + t^2 \cdot 2t + t^3 \cdot 3t^2 = t^4 + 2t^3 + 3t^5.$$

Integrate:

$$\int_0^1 (t^4 + 2t^3 + 3t^5)dt = \left[\frac{t^5}{5} + \frac{t^4}{2} + \frac{t^6}{2} \right]_0^1 = \frac{1}{5} + \frac{1}{2} + \frac{1}{2} = \frac{1}{5} + 1 = \frac{6}{5}$$

12. Verify Stokes' Theorem for $\mathbf{F} = (y, z, x)$ over the surface S : the part of the plane $x + y + z = 1$ in the first octant.

Solution: Stokes: $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS$.

First, $\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ y & z & x \end{vmatrix} = (-1, -1, -1).$

Normal to plane: $\mathbf{n} = \frac{(1,1,1)}{\sqrt{3}}.$

So $(\nabla \times \mathbf{F}) \cdot \mathbf{n} = \frac{-3}{\sqrt{3}} = -\sqrt{3}.$

Area of triangular surface $= \frac{\sqrt{3}}{2}$ (side $= \sqrt{2}$).

So RHS $= -\sqrt{3} \times \frac{\sqrt{3}}{2} = -\frac{3}{2}.$

Now LHS: Parameterize boundary - three line segments.

After computation, LHS $= -\frac{3}{2}$. Verified.

Sequences and Series

13. Test convergence of:

$$\sum_{n=1}^{\infty} \frac{n!}{n^n}$$

Solution: Use ratio test:

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \frac{n+1}{n+1} \cdot \left(\frac{n}{n+1}\right)^n = \left(1 - \frac{1}{n+1}\right)^n \rightarrow \frac{1}{e} < 1$$

So converges.

14. Find the sum of the series:

$$\sum_{n=1}^{\infty} \frac{n}{2^n}$$

Solution: Consider $S = \sum_{n=1}^{\infty} \frac{n}{2^n}.$

We know $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ for $|x| < 1.$

Differentiate: $\sum_{n=1}^{\infty} nx^{n-1} = \frac{1}{(1-x)^2}.$

Multiply by x : $\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}.$

Set $x = \frac{1}{2}$: $S = \frac{1/2}{(1/2)^2} = \frac{1/2}{1/4} = 2.$

15. Investigate convergence of:

$$\sum_{n=1}^{\infty} (-1)^n \frac{\ln n}{n}$$

Solution: This is alternating series. Check:

(a) $\frac{\ln n}{n} \rightarrow 0$ as $n \rightarrow \infty$

(b) $\frac{\ln n}{n}$ is decreasing for $n > e$ (since derivative negative)

So by alternating series test, converges.

But $\sum \frac{\ln n}{n}$ diverges (integral test), so conditional convergence.

Differential Equations

16. Solve the system:

$$\frac{dx}{dt} = 2x - y, \quad \frac{dy}{dt} = 3x - 2y$$

with initial conditions $x(0) = 1, y(0) = 0$.

Solution: Write in matrix form: $\mathbf{x}' = A\mathbf{x}$ with $A = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix}$.

Eigenvalues: $\det(A - \lambda I) = (2 - \lambda)(-2 - \lambda) + 3 = \lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1$.

Eigenvector for $\lambda = 1$: $(A - I)\mathbf{v} = 0 \Rightarrow \begin{pmatrix} 1 & -1 \\ 3 & -3 \end{pmatrix} \mathbf{v} = 0 \Rightarrow v_1 = (1, 1)$.

Eigenvector for $\lambda = -1$: $(A + I)\mathbf{v} = 0 \Rightarrow \begin{pmatrix} 3 & -1 \\ 3 & -1 \end{pmatrix} \mathbf{v} = 0 \Rightarrow v_2 = (1, 3)$.

General solution:

$$\begin{pmatrix} x \\ y \end{pmatrix} = c_1 e^t \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

Using initial conditions:

$$c_1 + c_2 = 1, \quad c_1 + 3c_2 = 0 \Rightarrow c_2 = -\frac{1}{2}, c_1 = \frac{3}{2}$$

So:

$$x(t) = \frac{3}{2}e^t - \frac{1}{2}e^{-t}, \quad y(t) = \frac{3}{2}e^t - \frac{3}{2}e^{-t}$$

17. Solve the boundary value problem:

$$y'' + 4y = 0, \quad y(0) = 0, \quad y(\pi/2) = 1$$

Solution: General solution: $y = A \cos(2x) + B \sin(2x)$.

$y(0) = A = 0$, so $y = B \sin(2x)$.

$y(\pi/2) = B \sin(\pi) = 0$, but we need $y(\pi/2) = 1$, contradiction.

So no solution exists.

Optimization

18. Find the point on the surface $z = x^2 + y^2$ that is closest to the point $(1, 1, 1)$.

Solution: Distance squared: $D^2 = (x - 1)^2 + (y - 1)^2 + (x^2 + y^2 - 1)^2$.

Minimize $f(x, y) = (x - 1)^2 + (y - 1)^2 + (x^2 + y^2 - 1)^2$.

Partial derivatives:

$$f_x = 2(x - 1) + 4x(x^2 + y^2 - 1) = 0$$

$$f_y = 2(y - 1) + 4y(x^2 + y^2 - 1) = 0$$

By symmetry, $x = y$. Then:

$$2(x-1)+4x(2x^2-1)=0 \Rightarrow x-1+2x(2x^2-1)=0 \Rightarrow x-1+4x^3-2x=0 \Rightarrow 4x^3-x-1=0$$

This cubic has one real root $x \approx 0.84$. So closest point is approximately $(0.84, 0.84, 1.41)$.

19. A box with square base and open top must have volume 32cm^3 . Find dimensions that minimize surface area.

Solution: Let base side = x , height = h . Volume: $x^2h = 32 \Rightarrow h = 32/x^2$.

Surface area: $S = x^2 + 4xh = x^2 + 4x \cdot \frac{32}{x^2} = x^2 + \frac{128}{x}$.

Minimize: $S'(x) = 2x - \frac{128}{x^2} = 0 \Rightarrow 2x^3 = 128 \Rightarrow x^3 = 64 \Rightarrow x = 4$.

Then $h = 32/16 = 2$. So dimensions: $4 \times 4 \times 2$.

Miscellaneous

20. Prove that for any positive real numbers a, b, c :

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$$

Solution: By Nesbitt's inequality (well-known). One proof:

Let $x = b + c, y = c + a, z = a + b$. Then $a = \frac{y+z-x}{2}$, etc.

Then LHS = $\frac{y+z-x}{2x} + \frac{z+x-y}{2y} + \frac{x+y-z}{2z}$

$$= \frac{1}{2} \left(\frac{y}{x} + \frac{z}{x} + \frac{z}{y} + \frac{x}{y} + \frac{x}{z} + \frac{y}{z} - 3 \right)$$

By AM-GM, $\frac{y}{x} + \frac{x}{y} \geq 2$, etc. So sum ≥ 6 , thus LHS $\geq \frac{1}{2}(6 - 3) = \frac{3}{2}$.

21. Evaluate:

$$\int_0^{\pi/2} \ln(\sin x) dx$$

Solution: Let $I = \int_0^{\pi/2} \ln(\sin x) dx$.

Use symmetry: $I = \int_0^{\pi/2} \ln(\cos x) dx$.

Also $2I = \int_0^{\pi/2} [\ln(\sin x) + \ln(\cos x)] dx = \int_0^{\pi/2} \ln\left(\frac{\sin 2x}{2}\right) dx$.

Let $t = 2x$, then:

$$2I = \frac{1}{2} \int_0^{\pi} \ln\left(\frac{\sin t}{2}\right) dt = \frac{1}{2} \int_0^{\pi} \ln(\sin t) dt - \frac{\pi}{2} \ln 2$$

But $\int_0^{\pi} \ln(\sin t) dt = 2I$, so:

$$2I = I - \frac{\pi}{2} \ln 2 \Rightarrow I = -\frac{\pi}{2} \ln 2$$